

CECILIA M. MARTINEZ RODRIGUEZ

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| [HTTPS://CECILIAMRTZ.GITHUB.IO/PORTFOLIO/](https://ceciliammrtz.github.io/Portfolio/) |

Mechanical Engineering and Physics student with strong coursework in heat transfer, fluid mechanics, materials, and energy systems. Experienced in technical research, instrumentation, and engineering analysis. Skilled in Python, SolidWorks, ANSYS, and scientific computing. Seeking mechanical engineering industry experience and to contribute to innovative energy and sustainability projects.

EDUCATION

BARNARD COLLEGE OF COLUMBIA UNIVERSITY, New York, NY

B.A. in Physics, GPA 3.88, May 2026

COLUMBIA UNIVERSITY SCHOOL OF ENGINEERING AND APPLIED SCIENCE (SEAS)

MS in Mechanical Engineering, Expected May 2027

ENGINEERING & RESEARCH EXPERIENCE

Let's Share the Sun (LSTS) – Adjuntas, Puerto Rico,

July 2026 - Present

- Authored LSTS' inaugural Impact Report, communicating metrics and highlights to potential donors.
- Designed a branded Sheila's Hearts of Hope wooden heart keepsake for national campaign.
- Partnered with executive leadership to host and manage an immersive corporate volunteer visit, overseeing site logistics for off-grid solar and battery installations at community homes.

Building Energy Research Lab – Columbia University SEAS,

Summer 2025

- Integrate health outcomes as a quantifiable metrics in building decarbonization planning.
- Model HVAC filtration impacts on indoor air quality and energy performance.
- Apply building energy simulation tools to evaluate decarbonization strategies.

High Energy Astrophysics Research – Barnard College / VERITAS / CTA Group,

May 2023 -Present

- Conducted research on high-energy astrophysics under Prof. Reshmi Mukherjee in the VERITAS/CTA group, specializing on the analysis of the blazar, B2 1811+31.
 - Operated and calibrated VERITAS telescope system during observational shift at VERITAS facilities in Arizona in April 2025.
 - Manuscript on B2 1811+31 flare event in progress.
 - Presented findings at Lida Orzeck '68 SRI (2023, 2024, 2025); Astrofest (2023, 2024); and MARLAM (2024).
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TECHNICAL SKILLS & AWARDS

- **Programming:** Python; Command Line Interface; MATLAB.
 - **Engineering & Modeling:** SolidWorks; Ansys; Building energy modeling; HVAC filtration modeling.
 - **Physics & Instrumentation:** Telescope operation and calibration; high-energy astrophysics data analysis.
 - **Data & Research:** Scientific computing; large-dataset analysis; technical writing.
 - **Languages:** English and Spanish (fully bilingual and biliterate).
 - **Awards:** Dean's List (2022, 2023, 2024, 2025, 2026).
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PROFESSIONAL REFERENCES:

Reshmi Mukherjee, Ph.D. - Professor of Physics & Astronomy, Barnard College; Research Mentor High Energy Astrophysics, rmukherj@barnard.edu 212-854-5101.

Nancy Brennan-Jordan, Executive Director of Let's Share the Sun Foundation, 501-c-3 not-for-profit, nancy@letssharethesun.org 518-326-1104

RELEVANT COURSEWORK

- **Classical Mechanics:** Newtonian + Lagrangian methods.
- **Electricity and Magnetism:** fields, circuits, electrodynamic waves and optics.
- **Classical Waves:** principles of wave motion; mechanical, acoustic, and electromagnetic waves.
- **Quantum Physics:** wave-particle duality; uncertainty principle; Schrödinger equation.
- **Mathematics:** Calculus I-IV, Linear Algebra, Ordinary Differential Equations.
- **Engineering:** Electronics lab, Mechanics of Solids, Fluid Mechanics, Intro to Control Systems, Heat Transfer, Materials and Processes Manufacturing, Computer Graphics and Design, Decarbonizing Buildings Studio, and Alternative Energy Resources and Building Energy Modeling and Simulation, Intro to Atmospheric Science.